

PHYSICS EXAM REVIEW

Motion, Kinematics & Dynamics

10SCI Unit 5

Covers: Describing Motion | Acceleration | Newton's Laws
Includes: Formulas • Practice Problems • Data Tables • Graph Exercises

SESSION 1: Describing Motion

Part A — Formulas & Key Concepts

1. Speed

Speed is the distance an object travels per unit of time. It is a scalar (magnitude only).

$$s = d / t$$

Variables

s = speed
d = distance
t = time

SI Units

m/s (or km/h, mi/hr, etc.)
metres (m)
seconds (s)

Relationship: Speed increases as distance increases (for a fixed time), or as time decreases (for a fixed distance).

Note: Speed is scalar — it has no direction. Common units: m/s, km/h, mi/hr, ft/s.

2. Average Speed

Average speed = total distance traveled ÷ total time taken.

$$s_{\text{avg}} = d_{\text{total}} / t_{\text{total}}$$

Variables

s_avg = average speed
d_total = total distance
t_total = total time

SI Units

m/s
metres (m)
seconds (s)

Relationship: Average speed does NOT depend on direction — it uses total path length, not displacement.

Note: A trip of 300 km in 5 hours → average speed = 60 km/h, even with stops and speed changes.

3. Velocity

Velocity is the speed of an object AND its direction of motion. It is a vector quantity.

$$\mathbf{v} = \Delta \mathbf{x} / t$$

Variables

v = velocity
 Δx = displacement (change in position)
t = time

SI Units

m/s (with direction)
metres (m)
seconds (s)

Relationship: Velocity = displacement ÷ time. Speed can remain constant while velocity changes if direction changes.

Note: Always include direction in your answer! e.g., 50 km/h North.

4. Displacement vs. Distance

Distance (d): Total path length traveled — scalar, always positive.

Displacement (Δx): Straight-line change in position from start to end — vector, can be negative.

Distance	Displacement
• Scalar (no direction) • Total path length • Always positive or zero • Measured in metres	• Vector (has direction) • Shortest path start→end • Can be negative or zero • Use Pythagorean theorem for 2D

5. Pythagorean Theorem (for 2D Displacement)

$c^2 = a^2 + b^2 \rightarrow c = \sqrt{a^2 + b^2}$	
Variables c = hypotenuse (displacement) a = horizontal distance b = vertical distance	SI Units metres (m) metres (m) metres (m)
Relationship: Used when an object moves in two perpendicular directions. The hypotenuse IS the displacement magnitude. Note: Only needed when path has a 90° turn. Always state direction as well (e.g., 10.2 blocks East).	

6. Slope on a Distance-Time Graph

On a distance vs. time graph, the slope of the line equals the object's speed.

$\text{slope} = \text{rise} / \text{run} = \Delta y / \Delta x = \text{distance} / \text{time} = \text{speed}$	
Variables rise = change in distance (Δd) run = change in time (Δt) slope = speed	SI Units metres (m) seconds (s) m/s
Relationship: A steeper slope = faster speed. A horizontal line = object is NOT moving. A curved line = changing speed.	

Graph Shapes — Distance vs. Time

Graph Shape	What It Means	Speed
Horizontal line	Object is NOT moving (stopped)	0 m/s
Straight diagonal line (up)	Moving away from reference at CONSTANT speed	Constant, positive

Straight diagonal line (down)	Moving toward reference at CONSTANT speed	Constant, positive
Curve (steepening)	Object is SPEEDING UP (accelerating)	Increasing
Curve (flattening)	Object is SLOWING DOWN (decelerating)	Decreasing

Part B — Practice Exercises

SECTION 1-1: Distance & Displacement

Solve the following problems. Draw a diagram before solving.

1. A cross-country team runs a 10 km race. They end up outside the locker room where they started. What is their distance? What is their displacement?

Distance: _____

Displacement: _____

2. Nicole parks her car at Target and walks 150 ft north to get to the store. On her way back to her car, she walks 50 ft before pausing. Find her distance and displacement.

Distance: _____

Displacement: _____

3. Maria walks 20 ft north, 35 ft east, and then 20 ft south to reach the pool. Find her distance and displacement.

Distance: _____

Displacement: _____

4. A delivery truck drives 18 blocks north, 10 blocks east, and 16 blocks south. All blocks are equal length. Find the distance and displacement. (Hint: use Pythagorean theorem for displacement!)

Distance: _____

Displacement working:

$$a^2 + b^2 = c^2 \rightarrow \quad^2 + \quad^2 = \quad^2$$

Displacement: _____

SECTION 1-2: Speed & Velocity

5. A student runs 250 m in 30 s. What is her average speed?

Formula used: _____

Working: _____

Answer: _____

6. Sound travels at 343 m/s. If a lightning bolt strikes 2 km away, how long does it take the sound to reach you?

Formula used: _____

Working: _____

Answer: _____

7. A space shuttle orbits Earth at 21,000 km/hr. How far will it travel in 5 hours?

Working: _____

Answer: _____

8. An elevator travels 210 m upward in 35 s. What is its velocity?

Working: _____

Answer (include direction): _____

SECTION 1-3: Graph Interpretation

Use the data tables below to create distance vs. time graphs, then answer the questions.

Data Table A

Time (s)
0
1
2
3
4
5

Data Table B

Time (s)
0
1
2
3
4
5

Graph A: Distance vs. Time

[Graph Space]

x-axis: Time (s) | y-axis: Distance (m)

Graph B: Distance vs. Time

[Graph Space]

x-axis: Time (s) | y-axis: Distance (m)

9. Describe the motion shown in Graph A. What is the object doing?

10. Calculate the speed in Graph B using the slope formula.

slope = rise/run = Δ distance/ Δ time = ____ / ____ = ____ m/s

11. On the graph below, sketch: (a) an object not moving, (b) an object at constant speed, (c) an object speeding up.

Sketch Graph: Describe motion with labels

[Graph Space]

x-axis: Time (s) | y-axis: Distance (m)

SESSION 2: Acceleration

Part A — Formulas & Key Concepts

1. Acceleration

Acceleration is the rate of change of velocity over time. An object accelerates when it speeds up, slows down, OR changes direction.

$$a = (v_f - v_i) / t$$

Variables

a = acceleration
v_f = final velocity
v_i = initial velocity
t = time

SI Units

m/s² (or km/h², mi/hr²)
m/s
m/s
seconds (s)

Relationship: Positive acceleration = speeding up. Negative acceleration (deceleration) = slowing down.

Note: Also written as $a = \Delta v / t$ where Δv = change in velocity.

2. Kinematic Equation 1 — Final Velocity

$$v_f = v_i + a \cdot t$$

Variables

v_f = final velocity
v_i = initial velocity
a = acceleration
t = time

SI Units

m/s
m/s
m/s²
seconds (s)

Relationship: Use when you know initial velocity, acceleration, and time and want to find final velocity.

Note: Rearranged: $v_i = v_f - at$ | $a = (v_f - v_i) / t$ | $t = (v_f - v_i) / a$

3. Kinematic Equation 2 — Displacement with Acceleration

$$s = v_i \cdot t + \frac{1}{2} \cdot a \cdot t^2 \quad (+s_0 \text{ if not starting from } 0)$$

Variables

s = displacement
v_i = initial velocity
a = acceleration
t = time
s₀ = initial position (if not 0)

SI Units

metres (m)
m/s
m/s²
seconds (s)
metres (m)

Relationship: Use when you need to find displacement (distance traveled) given time, initial velocity, and acceleration.

Note: If starting from rest: $v_i = 0$, so $s = \frac{1}{2}at^2$. If position starts at 0: drop s_0 .

4. Kinematic Equation 3 — No Time Variable

$$v_f^2 = v_i^2 + 2 \cdot a \cdot s$$

Variables

v_f = final velocity
 v_i = initial velocity
 a = acceleration
 s = displacement

SI Units

m/s
m/s
m/s²
metres (m)

Relationship: Use when TIME is not given or not needed. Relates velocity, acceleration, and displacement directly.

Note: Also written as: $v^2 = u^2 + 2as$ or $v_f^2 = v_i^2 + 2a(s - s_0)$

5. Slope on a Velocity-Time Graph

On a velocity vs. time graph, the slope of the line equals the object's acceleration.

$$\text{slope} = \Delta v / \Delta t = (v_f - v_i) / t = \text{acceleration}$$

Variables

Δv = change in velocity
 Δt = change in time
slope = acceleration

SI Units

m/s
seconds (s)
m/s²

Relationship: Positive slope = positive acceleration (speeding up). Negative slope = negative acceleration (slowing down). Horizontal line = zero acceleration (constant velocity).

Graph Shapes — Velocity vs. Time

Graph Shape	What It Means	Acceleration
Horizontal line at $v=0$	Object is NOT moving	0
Horizontal line ($v > 0$)	Moving at CONSTANT speed	0 (zero acceleration)
Straight line sloping UP	Object is SPEEDING UP	Positive (+)
Straight line sloping DOWN	Object is SLOWING DOWN	Negative (-)
Steep slope	Large acceleration	Greater $ a $

Part B — Practice Exercises

SECTION 2-1: Acceleration Calculations

12. A car accelerates from a standstill to 60 km/s in 10 s. What is its acceleration?

Formula: _____

Working: _____

Answer: _____

13. A jet is traveling at 80 m/s when it lands and parks in 10 s. What is its acceleration?

Working: _____

Answer: _____

14. A skateboarder has an acceleration of 1.5 m/s^2 . Starting from rest, what speed will he reach after 2 s?

Formula: $v_f = v_i + a \cdot t$ _____

Working: _____

Answer: _____

15. A roller coaster starts at 4 m/s. After 3 seconds, its speed is 22 m/s. Find its acceleration.

Working: _____

Answer: _____

16. A car approaches a stop sign at 5 m/s and comes to a stop in 4 seconds. What was its acceleration?

Working: _____

Answer: _____

SECTION 2-2: Kinematic Equations

17. A Ferrari with an initial velocity of 10 m/s accelerates at 5 m/s^2 for 3 seconds. What is its final velocity?

Formula: $v_f = v_i + a \cdot t$

Working: _____

Answer: _____

18. A ball is dropped from rest and accelerates at 9.8 m/s^2 due to gravity. How far does it fall in 3 seconds?

Formula: $s = v_i \cdot t + \frac{1}{2} \cdot a \cdot t^2$ ($v_i = 0$ since dropped from rest)

Working: _____

Answer: _____

19. A car accelerates from 0 to 30 m/s over 200 m. What is its acceleration? (No time given — use Equation 3)

Formula: $v_f^2 = v_i^2 + 2 \cdot a \cdot s$

Working: _____

Answer: _____

SECTION 2-3: Velocity-Time Graphs

Use the data table below to create a velocity vs. time graph, then answer the questions.

Time (s)	Velocity (m/s)
0	0
1	3
2	6
3	9
4	12
5	15
6	15
7	15
8	10
9	5
10	0

Velocity vs. Time Graph

[Graph Space]

x-axis: Time (s) | y-axis: Velocity (m/s)

20. Describe what is happening between t=0 and t=5 seconds.

21. Calculate the acceleration between t=0 and t=5 s.

Working: _____

Answer: _____

22. What is happening between $t=5$ and $t=7$ seconds?

23. Calculate the acceleration between $t=8$ and $t=10$ s. Is it positive or negative?

Working: _____

Answer: _____

SECTION 2-4: Match the Graph

Match each description (A–D) to the correct velocity-time graph shape below. Then write the acceleration type.

Description	Graph Type	Acceleration Type
A. Object not moving	_____	_____
B. Object moving at constant speed	_____	_____
C. Object speeding up	_____	_____
D. Object slowing down	_____	_____

SESSION 3: Newton's Laws of Motion

Part A — Formulas & Key Concepts

Newton's 1st Law — Law of Inertia

"An object will move at a constant velocity until a net force acts on it."

Key points:

- An object AT REST stays at rest unless a net force acts on it.
- An object IN MOTION stays in motion (same speed & direction) unless a net force acts on it.
- Inertia: the tendency of an object to resist a change in motion.
- More mass = more inertia (harder to change its motion).

Newton's 2nd Law — $F = ma$

A net force acting on an object causes it to accelerate in the direction of that net force.

$$F = m \times a$$

Variables

F = net force
m = mass
a = acceleration

SI Units

Newtons (N) — where $1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$
kilograms (kg)
 m/s^2

Relationship: Force is directly proportional to mass AND acceleration. Greater mass requires greater force for same acceleration.

Note: Rearranged: $m = F/a$ | $a = F/m$ | $\text{Weight} = \text{mass} \times 9.8 \text{ m/s}^2$

Weight (Special Case of $F = ma$)

$$W = m \times g$$

Variables

W = weight (force of gravity)
m = mass
g = acceleration due to gravity

SI Units

Newtons (N)
kilograms (kg)
 9.8 m/s^2 (on Earth)

Relationship: Weight is the gravitational force on an object. Mass is the amount of matter — it does NOT change with location. Weight DOES change.

Note: On the Moon, $g \approx 1.6 \text{ m/s}^2$, so you weigh $\sim 1/6$ of your Earth weight.

Newton's 3rd Law — Action-Reaction

"Every action has an equal and opposite reaction."

- When object A exerts a force on object B, object B exerts an equal force on A in the opposite direction.
- Forces are equal in size but opposite in direction — they act on DIFFERENT objects.
- Example: You push down on a trampoline → trampoline pushes you UP with equal force.

Momentum

$$p = m \times v$$

Variables

p = momentum
m = mass
v = velocity

SI Units

kg·m/s
kilograms (kg)
m/s

Relationship: Momentum = mass in motion. A heavier or faster object has more momentum.

Note: All moving objects have momentum. Momentum is a vector — it has direction.

Law of Conservation of Momentum

$$p_{\text{before}} = p_{\text{after}}$$

In a collision, total momentum is conserved (never created or destroyed — only transferred).

Net Force & Balanced/Unbalanced Forces

Force Type	Definition	Result on Object
Balanced Forces	Equal in size, opposite in direction → net force = 0	No change in motion (stays still or constant speed)
Unbalanced Forces	Not equal — net force ≠ 0	Object accelerates in the direction of the larger force
Net Force	The combination of all forces acting on an object	Determines the acceleration via $F = ma$

Forces Quick Reference

Force	Symbol	Formula	Unit
Net Force	F_{net}	$F = ma$	Newtons (N)
Weight	W	$W = mg$ ($g = 9.8 \text{ m/s}^2$)	Newtons (N)
Momentum	p	$p = mv$	kg·m/s
Acceleration	a	$a = F/m = (v_f - v_i)/t$	m/s^2

Part B — Practice Exercises

SECTION 3-1: Newton's 1st Law & Inertia

24. Why is Newton's 1st Law also called the Law of Inertia?

25. Rank the following from LEAST to MOST inertia: Pencil | Textbook | Kindergartener | Elephant

Order (least → most): _____

26. A driver is texting while driving and collides with a parked car, transferring all the car's moving force. Which law explains this?

Newton's ____ Law because:

27. When you slam on your brakes, your body still flies forward. Explain this using Newton's 1st Law.

SECTION 3-2: Newton's 2nd Law ($F = ma$)

28. With what force will a car (3,000 kg) hit a tree if it has an acceleration of 2 m/s^2 ?

Working: _____

Answer: _____

29. What force is needed to make a 10 kg bowling ball accelerate at 3 m/s^2 ?

Working: _____

Answer: _____

30. What is the mass of a truck if it accelerates at 5 m/s^2 with a force of 14,000 N?

Working: _____

Answer: _____

31. How much does a 22.5 kg suitcase weigh (in Newtons)?

(Hint: use $W = m \times g$, where $g = 9.8 \text{ m/s}^2$)

Working: _____

Answer: _____

32. Find the net force on the stack of books below:

← $F = 3 \text{ N}$



Books

$F = 1 \text{ N}$ →

Net Force = _____

Direction: _____

SECTION 3-3: Newton's 3rd Law & Momentum

33. Find the momentum of a 1,300 kg car traveling at 28 m/s.

Working: _____

Answer: _____

34. A 15 kg object moving at 3 m/s transfers ALL its momentum to a 5 kg object. What is the velocity of the 5 kg object after the collision?

$$p_{\text{before}} = p_{\text{after}} \rightarrow m_1 v_1 = m_2 v_2$$

Working: _____

Answer: _____

35. A 9 kg ball moving at 5 m/s transfers all its momentum to a 3 kg ball. What is the velocity of the 3 kg ball after the collision?

Working: _____

Answer: _____

36. Name the Newton's Law at work in each scenario:

Scenario	Newton's Law #	Reason (brief)
Your dirty clothes stay on the floor all night.	_____	
A kicked soccer ball eventually stops due to grass.	_____	
Jumping on a trampoline causes you to fly up.	_____	
A motorcycle reaches 60 mi/hr faster than a truck.	_____	
A fireman feels pushed backward when the hose is on.	_____	
A baseball player hits a ball toward the outfield.	_____	

SECTION 3-4: Force Diagrams & Net Force

For each scenario, calculate the net force and state whether it is balanced or unbalanced.

37. Two students push a box in the same direction with 50 N each. A third student pushes the opposite direction with 50 N. What is the net force?

50 N $\rightarrow$ + 50 N $\rightarrow$ + $\leftarrow$ 50 N

Net Force = _____

Balanced or Unbalanced?

38. One force of 7 N pushes right, another of 3 N pushes left on a box.

Net Force = _____

Direction of motion: _____

39. A 1,600 kg automobile needs a net force to give it an acceleration of 4.5 m/s^2 . What is the required net force?

Working: _____

Answer: _____

SESSION 4: Comprehensive Mixed Review

Full Formula Reference Sheet

Concept	Formula	Variables	Units
Speed	$s = d / t$	s=speed, d=distance, t=time	m/s
Velocity	$v = \Delta x / t$	v=velocity, Δx =displacement, t=time	m/s + direction
Average Speed	$s_{avg} = d_{total} / t_{total}$	Total distance ÷ total time	m/s
Acceleration	$a = (v_f - v_i) / t$	a=accel, v_f =final vel, v_i =initial vel	m/s ²
Kinematic 1	$v_f = v_i + a \cdot t$	Final velocity from initial + acceleration×time	m/s
Kinematic 2	$s = v_i \cdot t + \frac{1}{2} \cdot a \cdot t^2$	Displacement given time & acceleration	m
Kinematic 3	$v_f^2 = v_i^2 + 2 \cdot a \cdot s$	No time — velocity from displacement	m/s
Newton's 2nd	$F = m \times a$	F=force, m=mass, a=acceleration	N (kg·m/s ²)
Weight	$W = m \times g$	W=weight, m=mass, g=9.8 m/s ²	N
Momentum	$p = m \times v$	p=momentum, m=mass, v=velocity	kg·m/s
Conservation	$p_{1_before} = p_{2_after}$	Total momentum conserved in collision	kg·m/s
Pythagorean	$c = \sqrt{a^2 + b^2}$	For 2D displacement at 90° turn	m
D-T Graph slope	slope = speed	Rise/Run = $\Delta d / \Delta t$	m/s
V-T Graph slope	slope = acceleration	Rise/Run = $\Delta v / \Delta t$	m/s ²

Mixed Practice Problems

PART 1: Motion Word Problems

40. A school bus leaves school heading east for 2 miles, then turns north for 3 miles. Find the distance and displacement.

Distance: _____

Displacement (use Pythagorean): _____

Direction: _____

41. A supersonic jet flies 10 miles in 0.008 hours. Find the speed in mi/hr and m/s.

(1 mile = 1609 m)

Speed in mi/hr: _____

Speed in m/s: _____

42. An ostrich runs at 43 mi/hr. How far can it run in 15 minutes?

(Convert 15 min to hours first!)

Working: _____

Answer: _____

43. A helicopter's speed decreases from 60 m/s to 25 m/s in 5 seconds. What is its acceleration?

Working: _____

Answer: _____

44. A cart rolls down an incline from 2 m/s to 22 m/s with an acceleration of 4.0 m/s^2 . How long did it take?

Rearrange: $t = (v_f - v_i) / a$

Working: _____

Answer: _____

45. A bike takes 6 s to accelerate from 10 m/s to 14 m/s. The combined mass of bike and rider is 90 kg. Find the net force AND the final momentum.

Acceleration: _____

Net Force ($F=ma$): _____

Final Momentum ($p=mv$): _____

PART 2: Graph Analysis

Use the data tables below to plot TWO graphs (Distance vs. Time and Velocity vs. Time), then answer the analysis questions.

Mixed Data Table (Runner)

Graph Space — Distance vs. Time

Time (s)	Distance (m)
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

[Graph Space]

x-axis: Time (s) | y-axis: Distance (m)

Graph Space — Velocity vs. Time

[Graph Space]

x-axis: Time (s) | y-axis: Velocity (m/s)

46. During which time interval is the runner accelerating? How can you tell from the velocity-time graph?

47. During which time interval is the runner moving at constant speed?

48. Calculate the runner's acceleration during the first 5 seconds.

Working: _____

Answer: _____

49. What is the total distance traveled? What is the displacement if the runner ends up at 40 m from the start?

Total distance (from table): _____

Displacement: _____

PART 3: Newton's Laws Application Scenarios

Identify which Newton's Law applies AND predict the motion:

Scenario	Which Law?	Predict the Motion
You leave a cookie on a plate. An hour later it is still there.		
According to the Law of Universal Gravitation, you drop a tennis ball and an apple. What do you observe?		
You hit a ping-pong ball and a tennis ball with the same racket. Which has greater acceleration?		
You apply the force of your body down onto a trampoline. What does the trampoline do to you?		
A truck has a mass of 100 kg and velocity of 35 m/s. What is its momentum?	$p = mv$	
In a collision, a 9 kg ball at 5 m/s hits a 3 kg ball at rest. Find final velocity of 3 kg ball.	$p_1 = p_2$	

PART 4: Fill in the Blank — Key Vocabulary

Complete the paragraph using your notes:

Motion occurs anytime an object changes its 1. _____. This is entirely dependent on the 2. _____ point. Motion can be measured by distance, displacement, 3. _____, velocity, and acceleration. Velocity refers to an object's 4. _____ and direction. Acceleration refers to the rate of change in an object's 5. _____ over time. On a distance vs. time graph, the slope equals the object's 6. _____. On a velocity vs. time graph, the slope equals the object's 7. _____. When an object speeds up, it has 8. _____ acceleration. When it slows down, it has 9. _____ acceleration. Newton's 1st Law states that objects resist change in motion due to 10. _____. Momentum is 11. _____ in a collision and is calculated as 12. _____.

ANSWER KEY — Selected Problems

Session 1

Problem	Answer
S1 #5: Student 250m in 30s	$s = 250/30 = 8.33 \text{ m/s}$
S1 #6: Sound 2km at 343 m/s	$t = 2000/343 = 5.83 \text{ s}$
S1 #7: Space shuttle 5 hrs	$d = 21000 \times 5 = 105,000 \text{ km}$
S1 #8: Elevator 210m in 35s	$v = 210/35 = 6 \text{ m/s upward}$

Session 2

Problem	Answer
S2 #1: 0 to 60 km/s in 10s	$a = 60/10 = 6 \text{ km/s}^2$
S2 #2: Jet 80 m/s stops in 10s	$a = (0-80)/10 = -8 \text{ m/s}^2$
S2 #3: Skateboarder 1.5 m/s^2 for 2s	$v_f = 0 + 1.5 \times 2 = 3 \text{ m/s}$
S2 #4: Roller coaster 4 → 22 m/s in 3s	$a = (22-4)/3 = 6 \text{ m/s}^2$

Session 3

Problem	Answer
S3 #1: Car $3000\text{kg} \times 2 \text{ m/s}^2$	$F = 3000 \times 2 = 6000 \text{ N}$
S3 #2: 10 kg bowl at 3 m/s^2	$F = 10 \times 3 = 30 \text{ N}$
S3 #3: Truck 14000N at 5 m/s^2	$m = 14000/5 = 2800 \text{ kg}$
S3 #4: Suitcase 22.5 kg weight	$W = 22.5 \times 9.8 = 220.5 \text{ N}$
S3 #5: Car 1300 kg at 28 m/s	$p = 1300 \times 28 = 36,400 \text{ kg} \cdot \text{m/s}$

Good luck on your exam!

Remember: Show all your work, include units, and state direction for vector quantities.